

The objective of this lab exercise is to measure and analyze the magnetic field created by a circular loop of current.

Using the Biot-Savart Law, we found that the magnetic field at the center of a circular loop of current is given by the expression:

$$B = \frac{k \pi I}{R} \quad \text{where } R \text{ is the radius of the loop.}$$

This expression is valid for a single loop of current, i.e. a single wire carrying current I , but the circular coil we will use in this experiment consists of $N = 200$ loops of wire, each carrying current I . In essence, each single loop of wire creates an amount of magnetic field given by the above expression, so the total magnetic field is the sum of the magnetic fields created by the 200 loops of wire. We can modify the expression to account for the number of turns by simply multiplying the right side by N :

$$B = \frac{k \pi N I}{R}$$

In Part A of this lab, we will test this expression by measuring B for several values of I . Since k , π , N , and R are constants, we should find that B is proportional to I , so that a graph of B vs. I results in a straight line with a slope of $k\pi N/R$.

Procedure, Part A

The first step, of course, is to prepare your Excel sheet. You can calculate the expected value of the slope before you measure any data. To calculate the expected value:

- Measure and record the value of R , in meters.
- The value of N is 200 (with no units)
- Use the alternate value of k , i.e. **$0.2 \mu\text{T}\cdot\text{m/A}$**

Set the magnetic field sensor so that it is exactly at the center of the current loop. For current values of 0.10, 0.20, 0.30, ..., 0.80, measure the corresponding magnetic field with the magnetic field sensor. For each measurement:

- Set the current, then turn off the power supply.
- Zero (i.e. "tare") the magnetic field sensor.
- Immediately turn the power supply back on, verify the value of the current.
- Record the reading of the magnetic field sensor.

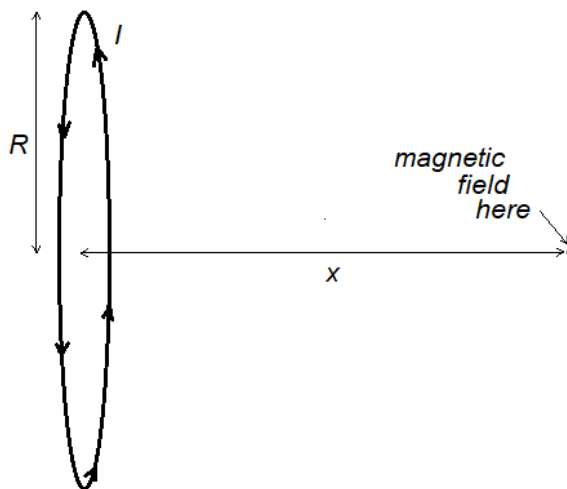
The reading of the magnetic field sensor will jump around quite a bit, as it is very sensitive. There is not much you can do about it; to take a reading, you may want to just take a picture... let your camera choose the value.

Note that your currents will be in **Amps** and your magnetic field will be in μT .

After recording the data, create a graph of B vs. I for the eight data points. You should find that the points line up well. Add the best-fit line and corresponding equation, and compare the measured value of the slope with your expected value.

Procedure, Part B

For Part B, we will consider the magnetic field a distance “ x ” along the axis of the circular loop of current. For R and x as defined in the diagram here:



The expression for the magnetic field at this point is:

$$B = \frac{k \pi I R^2}{(x^2 + R^2)^{3/2}}$$

We modify this expression by including a factor of N , for the same reason we did in Part A, and rearrange the expression algebraically to create a form that will be useful to us:

$$B^{-2/3} = \alpha x^2 + \alpha R^2 \quad \text{where} \quad \alpha = (k \pi N I R^2)^{-2/3}$$

It's not the prettiest equation, but it's all we've got. At least it allows us to create what should be a linear graph with our data by graphing $B^{-2/3}$ vs. x^2 .

As with Part A, set up your data table with a title and labels. You can calculate your expected value of the slope for Part B before measuring any data. We expect the slope of the graph to be “ α ”, from the equation above. And your expected value of “ α ” can be calculated using the values of k , N , I and R .

- For Part B, set the current at a value of 0.80 A.
- Be sure to use the standard version of k , $2 \times 10^{-7} \text{ T-m/A}$ (i.e. with standard units.)
- Calculate the expected value of the slope. (The result should be close to 10,000.)

The process to record data for Part B is very similar to that of Part A, except for Part B we will vary the distance “ x ”, from the center of the loop instead of varying the current.

- Set the magnetic field sensor exactly $x = 2.00 \text{ cm}$ from the center of the loop.
- Turn off the power supply and zero the magnetic field sensor.
- Turn on the power supply and record the reading of the magnetic field sensor.
- Repeat the above steps for values of 4.00, 8.00, 10.0, 12.0 and 14.0 cm.

- Create a data table for

x	in cm
B	in μT
x^2	in $\times 10^{-4} \text{ m}^2$
$B^{-2/3}$	in $\text{T}^{-2/3}$

- Create a graph of $B^{-2/3}$ vs x^2 . Add a best-fit straight line. Note that the units of the slope are $\text{T}^{-2/3} / \text{m}^2$.
- Compare the measured value of your slope with the expected value, i.e. calculate the percent difference.